This dataset gives the data of Income and money spent by the customers visiting a shopping mall. The data set contains Customer ID, Gender, Age, Annual Income, Spending Score. Therefore, as a mall owner you need to find the group of people who are the profitable customers for the mall owner. Apply at least two clustering algorithms (based on Spending Score) to find the group of customers.

Apply Data pre-processing Perform data-preparation (Train-Test Split) Apply Machine Learning Algorithm Evaluate Model.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.preprocessing import StandardScaler
```

```
df = pd.read_csv('/content/Mall_Customers.csv')

df.head()
```

	CustomerID	Genre	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

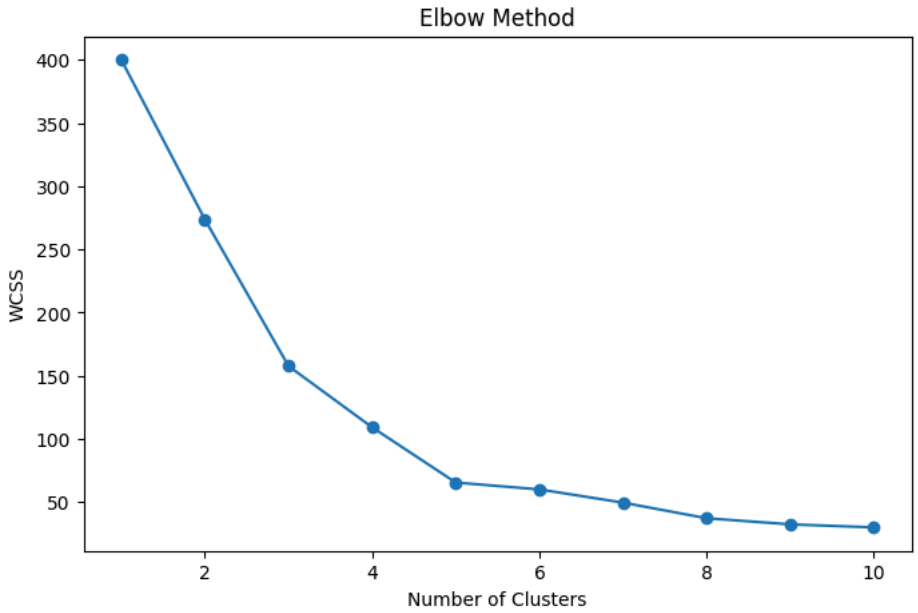
```
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]
```

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
wcss = []
k_values = list(range(1,11))

for k in k_values:
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

wcss_table = pd.DataFrame({
    "Number of Clusters (K)": k_values,
    "WCSS Value": wcss
})

print(wcss_table)
```

```
   Number of Clusters (K)  WCSS Value
0                       1  400.000000
1                       2  273.668887
2                       3  157.704008
3                       4  109.228227
4                       5   65.568408
5                       6   60.132875
6                       7   49.668245
7                       8   37.319123
8                       9   32.495081
9                      10   30.059323
```

```
plt.figure(figsize=(8,5))
plt.plot(range(1,11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()
```

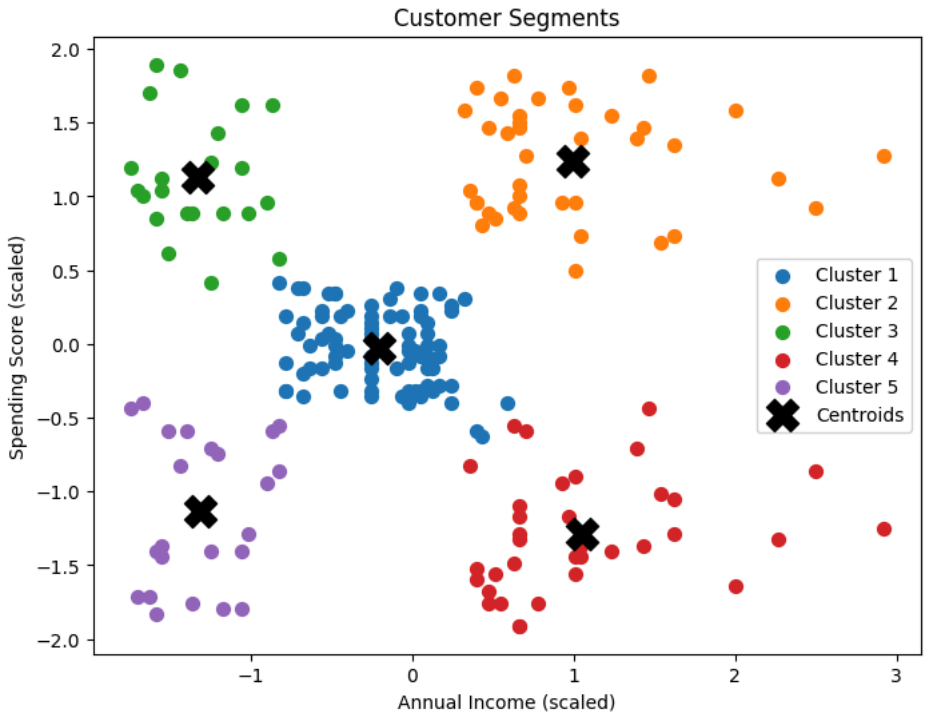



```
sil_scores = []
k_range = list(range(2,11))

for k in k_range:
    kmeans = KMeans(n_clusters=k, random_state=42)
    labels = kmeans.fit_predict(X_scaled)
    sil = silhouette_score(X_scaled, labels)
    sil_scores.append(sil)

sil_table = pd.DataFrame({
    "Number of Clusters (K)": k_range,
    "Silhouette Score": sil_scores
})

print(sil_table)
```

```
   Number of Clusters (K)  Silhouette Score
0                       2          0.397327
1                       3          0.466585
2                       4          0.494350
3                       5          0.554657
4                       6          0.513826
5                       7          0.502001
6                       8          0.455011
7                       9          0.456662
8                      10          0.444760
```

```
kmeans = KMeans(n_clusters=5, random_state=42)
y_pred = kmeans.fit_predict(X_scaled)
```

```
plt.figure(figsize=(8,6))

plt.scatter(X_scaled[y_pred==0,0], X_scaled[y_pred==0,1], s=50, label="Cluster 1")
plt.scatter(X_scaled[y_pred==1,0], X_scaled[y_pred==1,1], s=50, label="Cluster 2")
plt.scatter(X_scaled[y_pred==2,0], X_scaled[y_pred==2,1], s=50, label="Cluster 3")
plt.scatter(X_scaled[y_pred==3,0], X_scaled[y_pred==3,1], s=50, label="Cluster 4")
plt.scatter(X_scaled[y_pred==4,0], X_scaled[y_pred==4,1], s=50, label="Cluster 5")

plt.scatter(kmeans.cluster_centers_[:,0],
            kmeans.cluster_centers_[:,1],
            s=300, c='black', marker='X', label="Centroids")

plt.title("Customer Segments")
plt.xlabel("Annual Income (scaled)")
plt.ylabel("Spending Score (scaled)")
plt.legend()
plt.show()
```


Customer Segments
Spending Score (scaled)
Annual Income (scaled)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−1
0
1
2
3
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Centroids